Ruhan Bhavsar

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Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science - Mechatronics Engineering — *Expected 2028*

Experience

Embedded Systems Validation Developer (Motor Control) — Sept 2026 - Dec 2026
onsemi — *Waterloo, ON*

Engineering Automation Intern — Jan 2026 - Apr 2026
MPI Morheat — *Toronto, ON*

- Developed a low-cost automated **three-axis label cutter** for under **$500** using **stepper motors**, **3D-printed components**, and an **STM32** programmed in **C++**, saving **5–10 hours** of manual labor per week.
- Designed and fabricated an aluminum plate-bending roller with **welded components** and **geared rollers**, reducing heater-plate forming time from **20 minutes to 5 minutes per plate**.

Engineering Intern — May 2025 - Aug 2025
Vexos — *Markham, ON*

- Redesigned an **SMT rework nozzle** in **SolidWorks** and **SolidWorks PDM** for local machining, reducing supplier lead time from **6 weeks to 1 week** and per-unit cost by approximately **$700**.
- Designed and **3D printed** custom conformal-coating caps in **SolidWorks** to protect PCB components, improving process repeatability and supporting higher quality-control pass rates.
- Engineered an **automated curing fixture** for optical lens assemblies, eliminating **1.5 minutes** of operator supervision per board and saving several hours weekly.

Firmware Developer — Sept 2024 - Present
Midnight Sun Solar Car Team — *Waterloo, ON*

- Maintained embedded firmware for **8 vehicle microcontrollers**, managing **ADC sampling** and **real-time tasks** with a **10 ms latency target** for system synchronization.
- Built a **Pytest-based Hardware-in-the-Loop testing framework** to validate firmware on vehicle hardware, reducing manual testing time by **80%**.

Projects

Robotic Hand | *SolidWorks, 3D Printing, Python, MediaPipe, ESP32*

- Designed, modeled, and **3D printed** a **six-finger robotic hand** with servo-driven actuation, integrating mechanical components and embedded control for motion replication.
- Developed a **Python** and **MediaPipe** control system with **45 ms end-to-end latency**, transmitting finger states to an **ESP32** for real-time servo actuation.

Rescue Robot | *SolidWorks, AutoCAD, C++, RobotC*

- Designed and modeled the robot chassis, mechanical components, and pickup/drop-off mechanisms in **SolidWorks** and **AutoCAD** for autonomous rescue-supply delivery.
- Programmed a **LEGO EV3** robot in **RobotC** to autonomously navigate a grid using a Depth-First Search pathfinding algorithm.

Technical Skills

Mechanical Design & Prototyping: SolidWorks, SolidWorks PDM, AutoCAD, Bambu Studio, FDM 3D Printing, Fixture Design, Rapid Prototyping, Technical Drawings
Manufacturing & Validation: Manufacturing Automation, Mechanical Assembly, Test & Validation, Calibration, HIL Testing
Embedded Systems: STM32, ESP32, Arduino, ADCs, Servo Motors, Stepper Motors
Programming & Analysis: Python, MATLAB, C++, C, Excel, Git